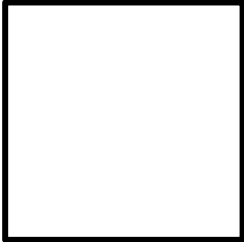


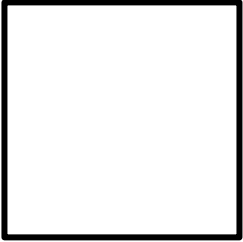


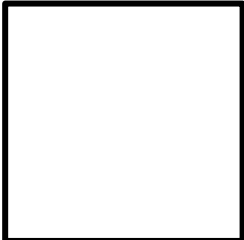
More complex binomial products

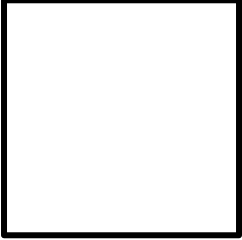
Task A: Using equivalent binomials

1) Use the area models to expand and simplify these expressions.

a)  $(y + 4)(7 - y)$
 $\equiv$ _____

c)  $(15 - y)(y + 5)$
 $\equiv$ _____

b)  $(-8 + y)(y - 8)$
 $\equiv$ _____

d)  $(x - y)(-y - 10)$
 $\equiv$ _____

2) Which of these expressions can be written as the difference of two squares.

a) $(x + 9)(9 - x)$

d) $(9 + x)(x - 9)$

g) $(-x + y)(y - x)$

b) $(x - 9)(9 - x)$

e) $(x + y)(y + x)$

h) $(x - y)(y + x)$

c) $(x + 9)(-x - 9)$

f) $(x + y)(-x + y)$

i) $(x - y)(y - x)$

3) Which products of two binomials give the expressions in the box?

$$x^2 - 81$$

$$81 - x^2$$

$$x^2 - y^2$$

$$y^2 - x^2$$



Task B: Coefficients other than 1

1) Expand and simplify each product of two binomials.

a) $(2x - 6)(x + 3)$

c) $(3x - 6)(2x + 6)$

e) $(3x - 6)(3x - 6)$

b) $(3x - 6)(x + 3)$

d) $(3x - 6)(2x - 6)$

f) $(3x - 6)(3x + 6)$

2) What do you notice about your answer to a) and f)? Can you explain why this happens.

3) Expand and simplify each product of two binomials.

a) $(2x + 3y)(4x + 5y)$

c) $(3y - 4x)(2x - 5y)$

e) $(-5y + 2x)(3y - 4x)$

b) $(2x + 5y)(4x + 3y)$

d) $(3x - 4y)(5y - 2x)$

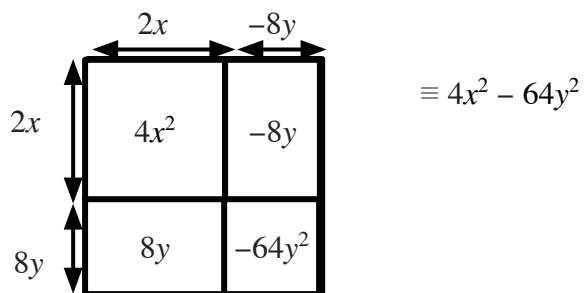
f) $(5y + 2x)(-3y - 4x)$

4) What do you notice about your answer to c) and e)? Can you explain why this happens.



5) Izzy wants to find the expanded form of the product $(2x - 8y)^2$

a) Spot and correct any mistakes she has made.



6) Can you write two binomials which will have the product $4x^2 - 64y^2$

Is there more than one way?

